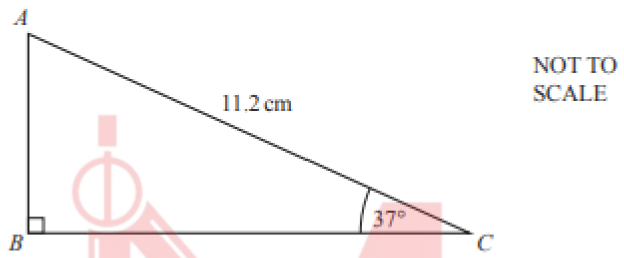
Trigonometry

Worksheet 1a

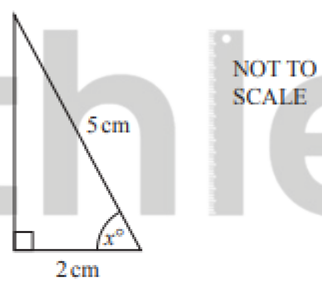
Basic, 90 Degree only

1. S15/q11/q11



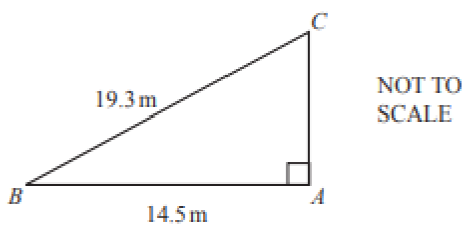
Calculate AB. [2]

2. S15/q13/q7



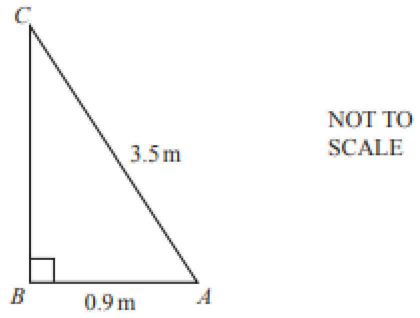
Calculate the value of x. [2]

3. W15/q12/q10



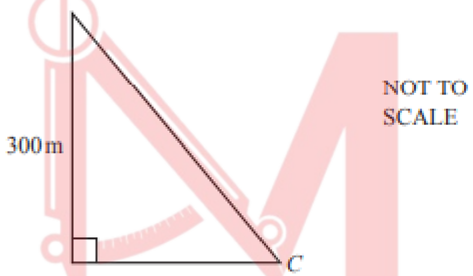
Use trigonometry to calculate angle ACB. [2]

4. M16/q22/q3



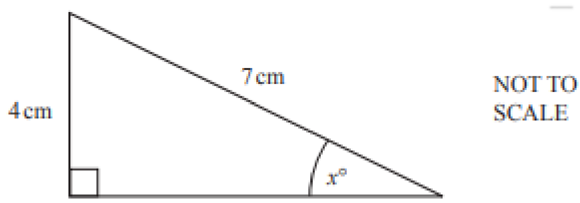
Calculate angle BAC. [2]

5. W16/q22/q9



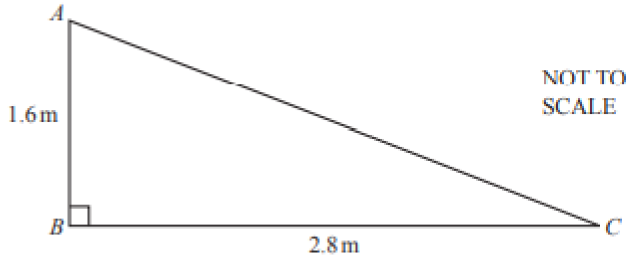
From the top of a building, 300 metres high, the angle of depression of a car, C, is 52°. Calculate the horizontal distance from the car to the base of the building. [3]

6. S17/q22/q12



Calculate the value of x. [2]

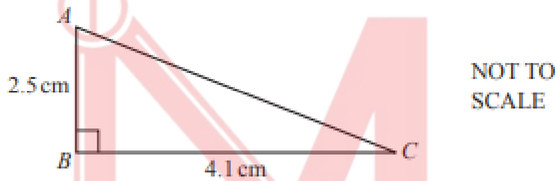
7. M18/q22/q16



a. Find the area of triangle ABC. [2]

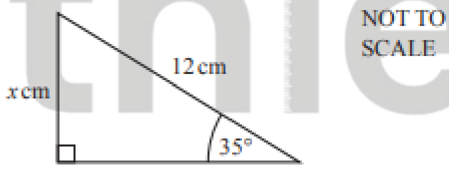
b. Calculate AC. [2]

8. S18/q21/q7



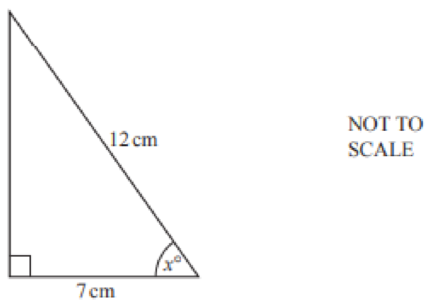
Calculate the length of AC. [2]

9. S19/q21/q7



The diagram shows a right-angled triangle. Calculate the value of x. [2]

10. W21/q21/q9



Calculate the value of x. [2]

Answer Key

1. S15/q11/q11

11	6.74[0…]	**2**	**M1** for $\frac{AB}{11.2} = \sin 37$ or better

2. S15/q13/q7

7	66.4[2......]	**2**	**M1** for cos [….=] $\frac{2}{5}$ oe

3. W15/q12/q10

10	48.7 or 48.70….	**2**	**M1** for sin[=]$\frac{14.5}{19.3}$ oe

4. M16/q22/q3

3	75.1 or 75.09 to 75.10	**2**	**M1** for cos […=] $\frac{0.9}{3.5}$

5. W16/q22/q9

9	234 or 234.3 to 234.4	**3**	**M2** for [dist =]$\frac{300}{\tan 52}$ oe or **M1** for correct implicit trig statement allow **M1** if they use *their* 52 or *their* 38 provided it is marked on the diagram or **B1** for 52 or 38 correctly placed If zero scored, **SC1** for final answer 384

6. S17/q22/q12

12	34.8 or 34.84 to 34.85	**2**	**M1** for sin [=] $\frac{4}{7}$

7. M18/q22/q16

16(a)	2.24	**2**	**M1** for $0.5 \times 1.6 \times 2.8$
16(b)	3.22 or 3.224 to 3.225	**2**	**M1** for $[AC^2 =]1.6^2 + 2.8^2$

8. S18/q21/q7

7	4.8[0] or 4.802…	**2**	**M1** for$[AC^2 =]\,2.5^2 + 4.1^2$

9. S19/q21/q7

7	6.88 or 6.882 to 6.883	**2**	**M1** for $\sin 35$ [=] $\frac{x}{12}$ oe or better

10. W21/q21/q9

9	54.3 or 54.31…	2	**M1** for $\cos[x] = \frac{7}{12}$ oe